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Models and Scenarios for Energy Planning

Endogenous Modal Shift of Road Freight in TEMOA-Italy

Database Development, Modal-Shift Architecture and Scenario Results

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Abstract

Can the EU freight modal-shift targets emerge from cost optimality, be bought with sector-internal incentives, or must they be mandated and at what price? This work answers the question for Italy by extending the freight sector of TEMOA-Italy, the Italian instance of the TEMOA (Tools for Energy Modeling Optimization and Analysis) energy system optimization model, with roughly 9,500 lines of SQL along two axes, the disaggregation of road-freight technologies into five distance classes (D1–D5, from urban hauls below 50 km to international hauls above 500 km) and the construction of an endogenous modal-shift layer on the long-haul (D5) market. The architecture (aggregated tonne-kilometre demand, zero-cost linkers, intermodal terminals with physical penalties, blended "shift-twin" technologies, disjoint constraint groups) is implemented once, in a single reference SQL patch (Part A + Part B), and reused unchanged across a 14-run scenario tree, two additional diagnostic runs forcing electric rail traction (FRZ_ELC), and 20 Modelling-to-Generate-Alternatives (MGA) iterations.

The policy answer is unambiguous. Rail modal shift saturates its EU ceiling with no incentive in every free run: it is a no-regret measure whose binding constraint is infrastructure, not economics. Maritime modal shift, by contrast, cannot be activated by any vessel-side financial incentive: the competitive gap , 668 M€ discounted in the Business-as-Usual scenario (BAU), 3,112 M€ in Net-Zero , exceeds the entire subsidizable chain cost (385 and 546 M€ respectively), because the barrier lies in the marine-fuel supply chain, not in the vessel. Forcing the maritime shift has an exactly measurable, (quasi-)linear price of 5.32 (BAU) and 14.95 (NT0) M€2020 per Gtkm·yr. In BAU the mandate is carbon-positive at every level of enforcement, since vessels fuelled by heavy fuel oil (HFO) displace the cleanest road vintages. Under the net-zero cap, the model concentrates residual freight emissions on diesel trains and grey ammonia synthesis , a rational linear-programming (LP) allocation under a system-wide constraint that the FRZ_ELC runs quantify and correct (freight emissions 3.20 → 2.37 Mt at 2050). A ten-sample Monte Carlo campaign on intermodal terminal costs (σ = 50%) leaves the modal allocation unchanged in every draw, certifying the rail and maritime findings as robust corner solutions rather than parameter-sensitive outcomes. Modal-shift targets for domestic navigation are therefore attainable only through regulatory mandates, at 0.67–3.1 bn€ discounted, or through upstream decarbonisation of marine fuels.

Nomenclature

Symbols, acronyms and scenario codes used throughout this report, listed here for reference so that no definition needs to be searched for at its first occurrence in the text.

Acronym	Definition
BAU	Business-as-Usual macro-scenario (no additional climate policy)
NT0	Net-Zero macro-scenario (system-wide CO2 cap + ETS2 carbon pricing)
ON / OFF	Modal-shift architecture present / absent in the database
FRZ (α)	Forced run: mandated maritime shift at α = 25/50/75/100% of the EU target
INC	Incentive run: subsidised vessel investment and fixed costs
FRZ_ELC 1/2	Diagnostic runs forcing electric traction on shift trains / whole rail sector
HTR / MTR / LCV	Heavy trucks / Medium trucks / Light Commercial Vehicles
D1–D5	Road-freight distance classes: <50 / 51–150 / 151–300 / 301–500 / >500 km
tkm, Gtkm	Tonne-kilometre; giga tonne-kilometre (freight service unit)
Bvkm	Billion vehicle-kilometres
LF	Load factor [t/vehicle]
TEMOA	Tools for Energy Modeling Optimization and Analysis
ESOM	Energy System Optimization Model
RES	Reference Energy System
LP	Linear Programming
MGA / HSJ	Modelling to Generate Alternatives / Hop-Skip-Jump objective formulation
GWP_100	Global Warming Potential over a 100-year horizon [kt CO2-eq]
CCUS	Carbon Capture, Utilisation and Storage
ETS2	EU Emissions Trading System for buildings and road transport (Dir. 2023/959)
SMR	Steam Methane Reforming
HFO / LNG / AMM	Heavy Fuel Oil / Liquefied Natural Gas / Ammonia (marine fuels)
DST, GSL, NGA, BIO	Powertrain codes: diesel, gasoline, natural gas, biofuel
PHEV, FULHYB	Plug-in hybrid, full-hybrid powertrains
ELC (BEV), FCELL (FCEV)	Battery-electric, hydrogen fuel-cell powertrains
TEU / Ro-Ro	Twenty-foot Equivalent Unit / Roll-on Roll-off (maritime logistics)
WACC	Weighted Average Cost of Capital
JRC-IDEES	JRC Integrated Database of the European Energy System
Output_*	TEMOA result tables: Output_Objective (total discounted cost), Output_VFlow_Out/In (commodity flows), Output_Costs, Output_Emissions

1. Introduction: the EU modal-shift policy

The EU's modal-shift policy is a core strategy to decarbonize transport by moving passengers and freight away from highly polluting modes (road and aviation) toward greener alternatives , rail, inland waterways and maritime transport , with climate neutrality by 2050 as the ultimate goal. The quantitative anchors are well established: 30% of long-haul road freight shifted to rail or waterborne modes by 2030 and more than 50% by 2050 (2011 Transport White Paper, COM(2011) 144); rail freight traffic +50% by 2030 and doubled by 2050, short-sea shipping +25% by 2030. (Sustainable and Smart Mobility Strategy, COM(2020) 789) [1], [2].

These targets, however, are quantities, not instruments. The question addressed here , with a purpose-built database , is whether the shift emerges spontaneously in a cost-optimal energy system, whether it can be bought with sector-internal incentives, or whether it must be mandated, and at what cost.

1.1 Case study: TEMOA-Italy

TEMOA-Italy is a model instance for the optimization of the Italian energy system, developed within an extended version of the TEMOA (Tools for Energy Modeling Optimization and Analysis) framework and maintained by the MAHTEP Group at the Department of Energy of Politecnico di Torino [3], [4]. As an energy system optimization model (ESOM), it applies linear programming (LP) to microeconomic theory under partial-equilibrium, competitive-market and perfect-foresight assumptions, minimizing the total discounted system cost (investment, fixed, variable and emission costs) in a technology-explicit, integrated representation of the whole energy system, structured as a Reference Energy System (Figure 1).

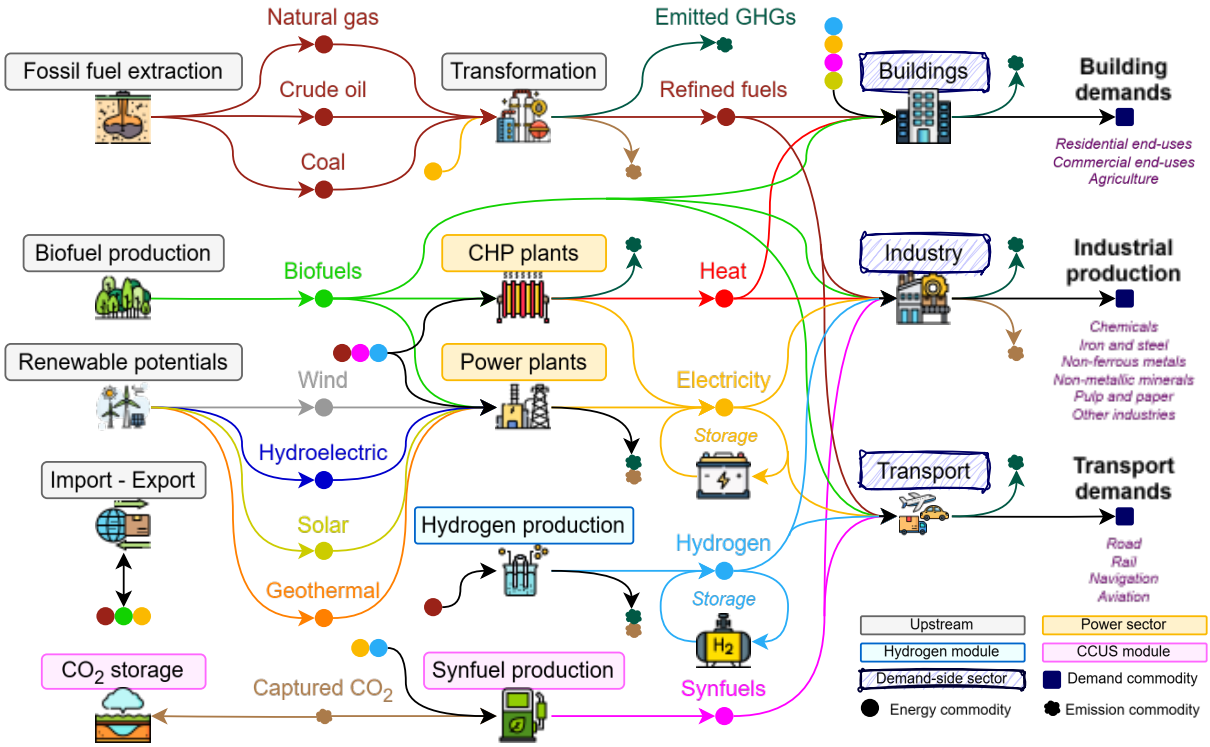


Figure 1: Reference Energy System of TEMOA-Italy: upstream supply, transformation, and end-use demand sectors.

The model adopts a time horizon extending to 2050, with the system evolution analyzed in five-year increments; it is built on IEA historical statistics for the base year 2006 and calibrated up to 2022 on Eurostat Energy Balances. For computationally heavy problems, a reduced milestone set (here 2030/2040/2050) is used.

Three properties of the underlying linear program matter directly for the results of this work.
Proportionality and additivity make the objective linear in any imposed activity level: the forcing ladder of

4.3.3 inherits its (quasi-)linearity from the LP structure itself, which is why exact linearity in BAU is a verification result and not a coincidence. The absence of integrality and of diminishing returns produces the well-known winner-takes-all behaviour on new installations: at equal service, the technology with the lowest total discounted cost captures the entire new vintage , the mechanism behind the diesel-rail allocation discussed in §4.4.4. Finally, duality guarantees that every binding constraint carries a shadow price, which is what makes the marginal cost of the mandate an exactly measurable quantity rather than an estimate.

Before this work, freight was represented by single aggregate road categories per weight class and by three independent exogenous demands (road; rail; navigation): no modal substitution was possible by construction, and the road mix could not reflect haul length. Trains and vessels were not modified in their internal structure by this project (at most, prices were updated); everything happens in the demand architecture built around them.

2. Methodology and database architecture

The reference patch , a single SQL file appended to the model database, identical in the BAU and NT0 sources (~9,460 lines; see the Data and code availability note) , is organised in two parts: Part A , distance-class disaggregation; Part B , the modal-shift patch. All modifications follow an INSERT-only discipline with self-documenting notes per row (source and formula in the notes field), so the database itself is the audit trail.

2.1 Part A , Distance-class disaggregation (D1–D5)

Each road-freight technology (heavy trucks HTR, medium trucks MTR, light commercial vehicles LCV; existing and new vintages; 7–9 powertrains) is replicated across five distance classes, generating ~350 new rows in the technologies table plus the full parameter set:

```
INSERT INTO "technologies" VALUES
('TRA_ROA_HTR_NGA_N_D5','p','TRA','Heavy trucks - Natural gas - New (>500 km)','');
```

2.1.1 Activity calibration and distance shares

Activity is calibrated on JRC-IDEES vehicle-kilometres [5] crossed with the Eurostat road_go_ta_dc distance-band statistics [6]. The resulting distance shares by vehicle category are reported in Table 1.

Table 1: Distance shares by vehicle category (Eurostat / ICCT+T&E / MAHTEP estimate).

Distance class	HGV (HTR+MTR) %	MTR %	LCV %	Operational profile
<50 km (D1)	5.6%	15.0%	37.0%	Urban delivery (last-mile, couriers, utilities)
51–150 km (D2)	17.9%	25.0%	32.5%	Regional distribution (retail, supplies)
151–300 km (D3)	27.7%	27.5%	20.5%	Inter-regional (hub-to-hub)
301–500 km (D4)	19.4%	17.5%	6.5%	Long haul (articulated only)
>500 km (D5)	29.4%	15.0%	3.5%	International (almost exclusively articulated)
Sum	100.0%	100.0%	100.0%	,

2.1.2 Load factors

Where necessary, powertrain efficiencies were defined by combining literature values with load factors assumed per distance class (JRC + Eurostat; LCV scaled, HTR direct, MTR = 0.55 × HTR per ICCT), Table 2.

Table 2: Load factors by distance class and vehicle category [t/veh].

Distance class	LF LCV	LF MTR	LF HTR	Typical driving cycle
<50 km	0.1836	7.515	13.663	Urban / last-mile
51–150 km	0.2054	8.408	15.287	Mixed regional
151–300 km	0.2271	9.299	16.908	Inter-regional (BASE)
301–500 km	0.2319	9.492	17.258	Long haul
>500 km	0.2230	9.128	16.596	International

2.1.3 Efficiencies and learning curves

Efficiencies are anchored energy-backward to the Eurostat energy balance (Bvkm/PJ = Vkm_JRC / Energy_Eurostat), so the fleet reproduces statistical fuel consumption by construction. Starting from the raw specific consumptions in MJ/tkm, the load factors convert to model units as: Efficiency [Bvkm/PJ] = 1 / (MJ/tkm × LF). Efficiency-improvement trajectories were assumed for new vintages (Table 3), with larger improvement factors for BEV, fuel-cell and PHEV powertrains, which are farther from their thermodynamic limits.

Table 3: Learning curves for specific consumption [MJ/tkm] and improvement factors.

Tech	Base 2020	2030	2050	F 20→30	F 30→50	F 20→50
HTR Diesel_N	0.8840	0.8133	0.7320	0.920	0.900	0.828
HTR BEV	0.3075	0.2614	0.2195	0.850	0.840	0.714
HTR FCEV	0.4324	0.3459	0.2629	0.800	0.760	0.608
MTR Diesel_N	1.5640	1.4076	1.2387	0.900	0.880	0.792
MTR BEV	0.5100	0.4335	0.3641	0.850	0.840	0.714
MTR FCEV	0.7650	0.6120	0.4651	0.800	0.760	0.608
LCV Diesel_N	14.3128	12.5953	11.0838	0.880	0.880	0.774
LCV BEV	4.0678	3.4577	2.9044	0.850	0.840	0.714
LCV FCEV	6.3278	5.0622	3.8473	0.800	0.760	0.608
LCV NGA_N	14.3128	13.1678	11.8510	0.920	0.900	0.828

2.1.4 Costs

CostVariable and CostInvest were collected from multiple sources and adapted to the distance classes: CostVariable [M€/Bvkm] = retail_markup [M€/PJ] × dist_factor / efficiency [Bvkm/PJ]; CostInvest [M€/(Bvkm/yr)] = vehicle_cost [€] × 1000 / annual_km_per_vehicle. New-vintage market anchors are documented in-line in the notes field (e.g. '2030: IEA GEO 2024 STEPS, EU truck fleet BEV ~8%; IT×0.80 = 6%' [7]). A unit fix was applied to rail fixed costs (M€/PJ → M€/Gtkm, = 20/eff_base), documented per row, a correction that makes the economic rail shift possible at all.

2.2 Part B , MS_update: the endogenous modal-shift architecture

2.2.1 One aggregated demand

To simulate the modal shift endogenously, the three long-haul (D5) freight demands, HTR_D5, freight rail and domestic navigation, are merged into a single commodity in Gtkm. The demand is not new data: it is derived from the exogenous Bvkm demands of the MS-OFF scenarios, converted to Gtkm through the modal load factors, each derived as the empirical payload ratio LF [t/veh] = reported tonne-km ÷ vehicle-km: ≈16 t/veh for HTR (Eurostat `road_go_ta_dc`, 2019–2023 mean; JRC-IDEES HGV cross-check 15.7), ≈393 t/train for rail (JRC-IDEES rail tonne-km ÷ train-km, 2021–2022 window; train-km independently confirmed by Eurostat `rail_tf_trainmv`), and ≈2,119 t/vessel for domestic navigation (JRC-IDEES domestic-coastal tonne-km ÷ vessel-km, 2019–2023 mean 2,142).making the three modes commensurable on a common tonne-kilometre basis:

```
INSERT INTO "commodities" VALUES ('TRA_FRG','d','Aggregated freight demand - Modal Shift [Gtkm]');
INSERT INTO "Demand" VALUES ('IT',2030,'TRA_FRG', 91.8756,'Gtkm','HTR_D5+RAIL+NAV via LF');
INSERT INTO "Demand" VALUES ('IT',2050,'TRA_FRG',116.5266,'Gtkm','HTR_D5+RAIL+NAV via LF');
```

2.2.2 Linkers

Three zero-cost pass-through technologies (Efficiency = 1 Gtkm/Gtkm, LifetimeTech = 100 y, dummy ExistingCapacity = 200 Gtkm) connect each mode's native output to TRA_FRG, without touching the modes themselves:

```
INSERT INTO "Efficiency" VALUES
('IT','TRA_ROA_HTR_D5','TRA_LNK_ROA_HTR_D5',2006,'TRA_FRG',1.0,'Gtkm/Gtkm');
```

2.2.3 Terminals and access commodities

Two shift technologies consume ethos (free source commodity) and produce the access commodities. They embody the intermodality penalty as an efficiency haircut $\eta = 1 - p$ (detour treated as a proportional loss): 0.87 for rail (p = 13%: route circuity + terminal transfer) and 0.78 for sea (p = 22%: coastal detour + port

dwell + Ro-Ro utilisation void). These penalty fractions are the authors' engineering estimates, built from component circuity and handling factors; the terminal infrastructure cost per tonne-kilometre is sourced separately in-line (AdSP 2023):

```
INSERT INTO "Efficiency" VALUES
('IT','ethos','TRA_MS_TRD5_NAV',2030,'TRA_NAV_ACCESS',0.78,'Gtkm/Gtkm detour+port+void Ro-Ro 22%');
INSERT INTO "CostInvest" VALUES
('IT','TRA_MS_TRD5_NAV',2007,9.2857,'M€/Gtkm','65 EUR/TEU / (14t × 500km avg haul AdSP 2023)');
INSERT INTO "DiscountRate" VALUES ('IT','TRA_MS_TRD5_NAV',2007,0.07,'port-infrastructure WACC');
```

A MaxActivity "valve" keeps the shift off until 2025 and non-binding afterwards (= EU ceiling / terminal efficiency), defined at every milestone. A practical modeling caveat is recorded here: any temporary cap must carry an explicit terminal-period value, otherwise the db-creator forward-extrapolates the last value, here the zero, and silently kills the mechanism. The binding cap remains at group level.

2.2.4 Shift twins

Nine twins (3 rail, 6 navigation) are copies of the ordinary technologies that consume two inputs, the mother technology's fuel and the access commodity, and produce the mother's output. Since TEMOA's multi-input flow rule is additive, the twin's Efficiency is a blending coefficient, identical on both inputs: eff_twin = 1 / (1 + e_base), with e_base the mother's physical intensity in PJ/Gtkm. TechInputSplit then fixes the input shares (fuel = 1 − eff_twin, access = eff_twin). Per 1 Gtkm of output, the twin thus draws exactly e_base PJ of fuel and 1 Gtkm of access, same energy, same emissions, same costs as the mother, plus the terminal chain. Example (HFO vessel, e_base = 2.7593 PJ/Gtkm → eff = 0.266007):

```
INSERT INTO "Efficiency" VALUES ('IT','TRA_HFO','TRA_NAV_DOM_HFO_SHIFT_N',2030,'TRA_NAV_DOM',
  0.266007,'calib: E_base=0.3624, PJ/Gtkm=2.7593');
INSERT INTO "Efficiency" VALUES ('IT','TRA_NAV_ACCESS','TRA_NAV_DOM_HFO_SHIFT_N',2030,'TRA_NAV_DOM',
  0.266007,'access calib 1:1');
INSERT INTO "TechInputSplit" VALUES ('IT',2030,'TRA_HFO','TRA_NAV_DOM_HFO_SHIFT_N',0.733993,'energy
split');
INSERT INTO "TechInputSplit" VALUES
('IT',2030,'TRA_NAV_ACCESS','TRA_NAV_DOM_HFO_SHIFT_N',0.266007,'access split');
```

One documented exclusion: the dual-fuel twin (HFO+MET, three inputs) is left in the code fully written but commented out, the additive split cannot guarantee the 50/50 fuel ratio together with the 1:1 access ratio, and keeping the dead block in the source records the decision.

2.2.5 Groups and constraints

Base and shift technologies live in disjoint groups. The base groups are frozen at the BAU floor (Min = Max), so the pre-existing modal traffic can neither be cannibalised nor double-counted; the shift groups carry only a ceiling (EU target − floor) and no minimum, the shift is an option, not an obligation. The terminals belong to no group (they produce ACCESS, not freight), so the caps count real tonne-kilometres only:

```
INSERT INTO "MinActivityGroup" VALUES ('IT',2030,'RAIL_BASE_GRP',34.574,'BAU floor');
INSERT INTO "MaxActivityGroup" VALUES ('IT',2030,'RAIL_BASE_GRP',34.574,'BAU floor');
INSERT INTO "MaxActivityGroup" VALUES ('IT',2030,'RAIL_SHIFT_GRP',26.706,'EU-floor');
INSERT INTO "MaxActivityGroup" VALUES ('IT',2030,'NAV_SHIFT_GRP', 2.167,'EU-floor');
```

Scenario levers touch only this layer: the forced runs add MinActivityGroup on NAV_SHIFT_GRP at α × ceiling; the incentive runs discount CostInvest/CostFixed of the twins (in multiplier form, e.g. 140.1361*0.35, so the discount is self-documenting); NT0 adds the CO2 cap and the ETS2 CostEmission rows [8]. The architecture never changes across the tree, one patch, fourteen runs.

3. Scenario tree

The simulation campaign is organised as a binary tree (Figure 2). At the first level, the modal-shift architecture is either absent (OFF) or present (ON); each branch is run under two macro-scenarios, Business-as-Usual (BAU) and Net-Zero (NT0, with system-wide CO2 cap and ETS2 carbon pricing). Within ON, the shift is either free (the solver decides, with or without financial incentives) or forced (FRZ: a minimum modal-shift obligation is imposed on the maritime technology only, since rail reaches its ceiling spontaneously). The forced branch is enforced at four percentage steps of the EU target: 25, 50, 75 and 100%.

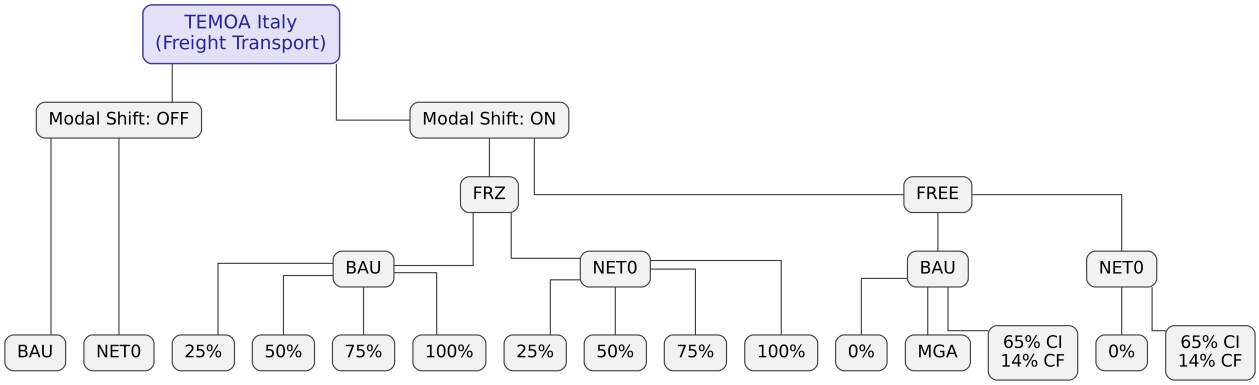


Figure 2: Scenario tree: OFF/ON architecture, BAU/NT0 macro-scenarios, free/incentive/forced branches with the α forcing ladder.

Table 4: 14 solved runs + 20 MGA iterations on one shared architecture.

Family	Runs	Lever (tables touched)
OFF	OFF_BAU, OFF_NT0	Architecture absent, reference
ON free	BAU_0, NT0_0	None
ON free + incentive	BAU_INC, NT0_INC	CostInvest/CostFixed on twins (+ CostEmission ETS2 in NT0)
ON forced	BAU/NT0_FRZ 25/50/75/100	MinActivityGroup NAV_SHIFT_GRP = α × ceiling
MGA	20 HSJ iterations on BAU_0, slack 1%	MGA objective, tech_mga

Run labels follow the pattern {scenario}_{lever}: e.g. BAU_0 = BAU, architecture ON, no lever; NT0_FRZ75 = Net-Zero with maritime mandate at 75% of the EU target.

Two additional diagnostic NT0 runs address rail traction (motivated in §4.4): NT0_FRZ_ELC_1, in which the shift trains are forced to be fully electric, and NT0_FRZ_ELC_2, in which the entire rail sector is forced to be electric.

Finally, a Modeling-to-Generate-Alternatives (MGA) analysis was performed on BAU_0 (no incentive). In MGA, the original objective (total system cost) becomes a constraint (cost ≤ optimum × (1 + slack)) and a new objective , here the activity-weighted Hop-Skip-Jump (HSJ) formulation, is minimized to generate maximally different near-optimal solutions. The purpose was to test whether vessel installation could appear within a near-optimal neighbourhood of the cost optimum.

Together, the three extensions probe orthogonal dimensions of the results: the ELC variants their allocation rationale, the MGA iterations their near-optimal neighbourhood, and the Monte Carlo campaign (§4.7) their parametric robustness. The scenario tree encodes decision alternatives; these layers test how solid each node is.

4. Results and analysis

4.1 Verification

Every solved database was audited on inputs (TechGroupWeight membership, access chains, terminal MaxActivity, live levers) and on the TEMOA output tables: Output_Objective (total discounted system cost), Output_VFlow_Out/In (commodity flows by technology), Output_Costs (cost breakdown). A convexity check on the NT0 forcing ladder confirmed the consistency of the objective values along the whole ladder, with the free Net-Zero baseline (Output_Objective = 4,896,336.2 M€) independently verified by re-solving the run.

4.2 Free runs vs. incentive runs (comparison C1)

Rail shift saturates its EU ceiling exactly in every free run, in both scenarios, with zero incentive; the maritime shift is zero everywhere (Figures 3–4). The incentive runs, a double incentive on vessel CostInvest and fuel CostFixed, are not merely similar to the free runs: they are identical on every extracted metric, period by period. Same objective to the decimal, same freight-sector GWP_100 (e.g. at 2050: 24,649.65 kt in BAU and 3,358.00 kt in NT0, identical in the free and incentive runs), same naval shift (zero). An incentive that changes nothing anywhere is the empirical signature of a lever applied to a technology the optimum never touches. Consequently, the scenario comparison concentrates on the four reference cases: BAU and NT0 economically free on the modal shift, and BAU and NT0 obliged to fully satisfy the EU policy (FRZ100).

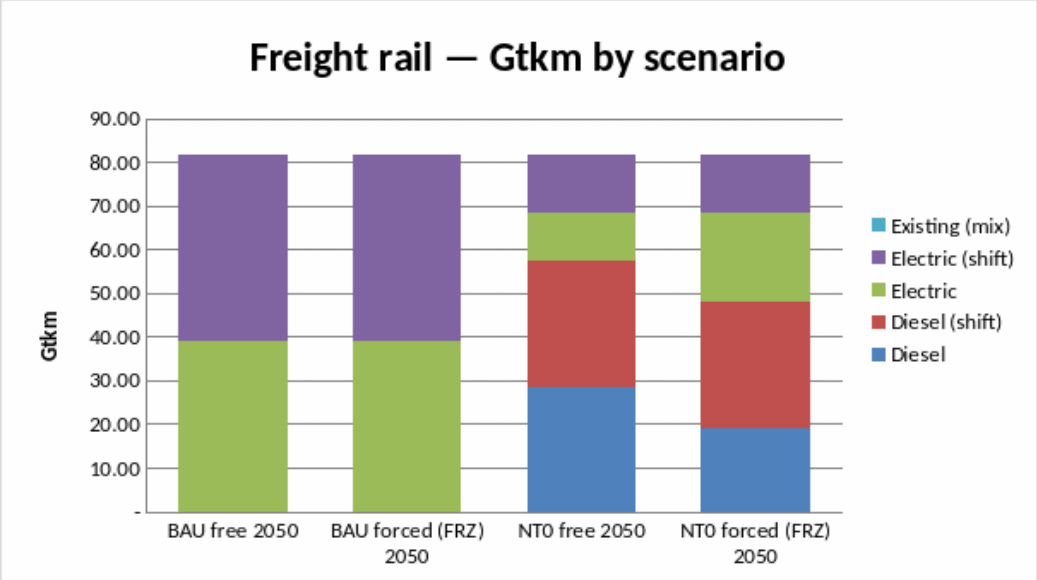


Figure 3: Rail freight activity [Gtkm] per scenario: the endogenous shift saturates the EU ceiling in every free run, split between diesel and electric traction (base and shift-twin technologies).

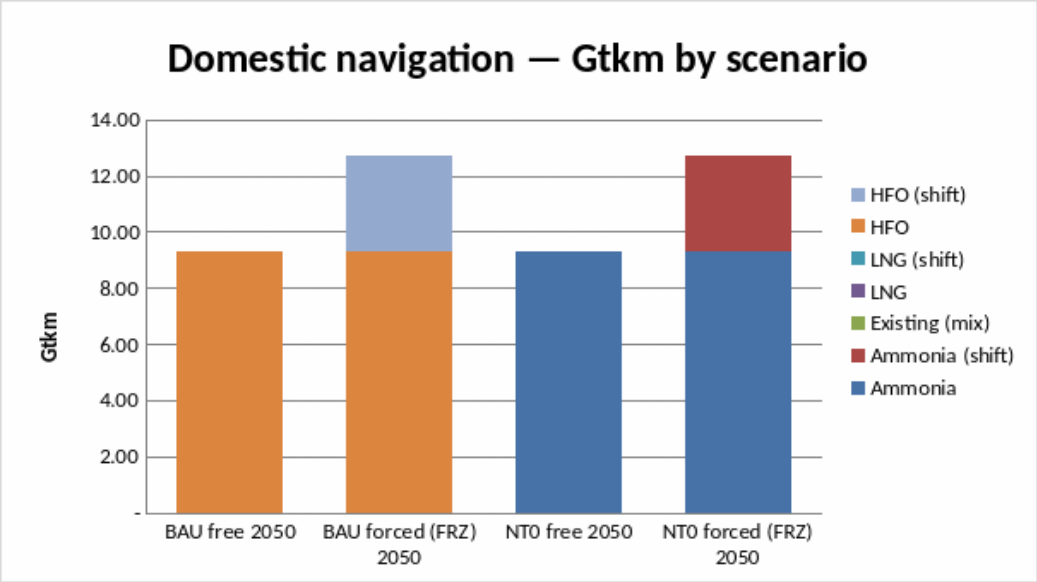


Figure 4: Domestic navigation activity [Gtkm] per scenario: the maritime shift is zero in every free run and appears only where mandated (FRZ), fuelled by HFO in BAU and by ammonia (with LNG bridge) in NT0.

4.3 Forced maritime shift and the accounting proof

4.3.1 Upstream view: where the energy goes

In BAU_FRZ100 versus free, the naval twins draw +23.0 PJ of HFO cumulatively (5.98/7.61/9.42 PJ in 2030/2040/2050), while the displaced trucks release only 3.4 PJ of gas and 1.3 PJ of diesel: the forced shift increases final energy use. Upstream, refinery throughput rises by +191 PJ and crude imports by +20.0 PJ, while gas imports fall by 3.2 PJ (commodity flows of the upstream sector, Output_VFlow_Out). The shift swaps cheap imported gas for expensive imported crude: the barrier lives in the fuel supply chain, not in the vessel.

4.3.2 The cost bridge (incentive infeasibility)

Decomposing Output_Costs of forced versus free runs partitions the forcing cost into the shift chain (A), the marine fuel chain (B), avoided road assets and fuels (C, D), and system rebalancing (E), Table 5 and Figure 5.

Table 5: Output_Costs bridge, forced minus free [discounted M€]. Gap > subsidizable base (block A) in both scenarios.

NT0 blocks are computed from Output_Costs (discounted, FRZ100 minus free) with explicit chain attribution: B aggregates ammonia/LNG/HFO delivery, grey-ammonia synthesis and its gas/H2/CCUS feed; E aggregates passenger-car re-electrification (+1,370.3) and re-abatement outside the transport sector (+2,207.1) under the common cap; a +28.5 M€ salvage/loan residual closes the sum to +3,112.5.

Cost bridge (discounted M€)	BAU	Net-Zero
A. Shift chain (vessel capex + O&M + terminal)	+384.9	+545.5
B. Marine fuel chain upstream	+739.5	+311.0
C+D. Avoided road assets & fuels	−523.4	−1,349.8
E. System rebalancing	+67.4	+3,577.3
Net gap ΔC	+668.4	+3,112.5

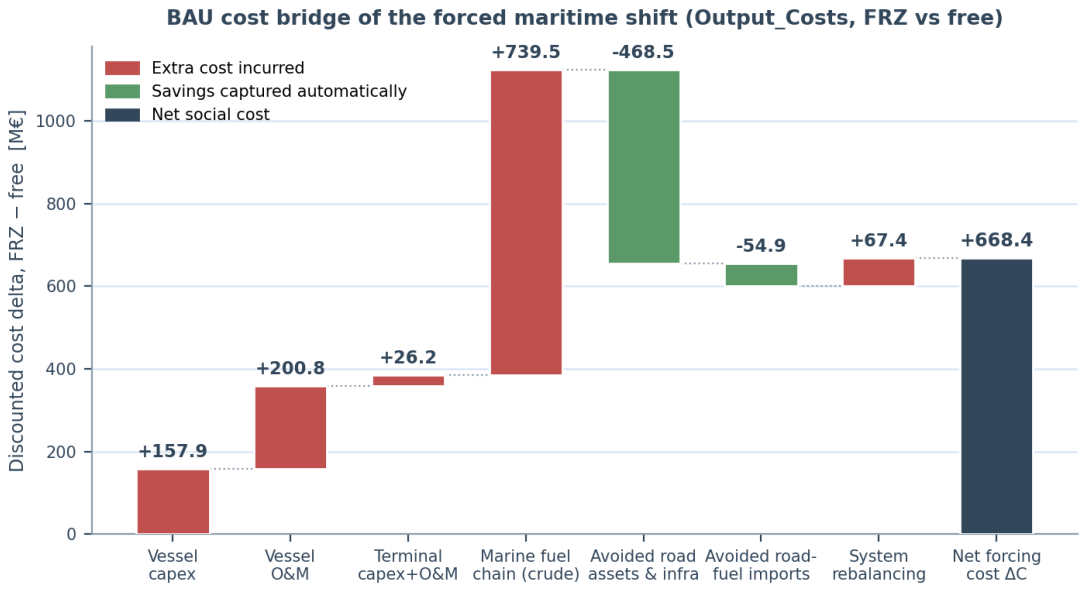


Figure 5: BAU cost bridge of the forced maritime shift (Output_Costs, FRZ100 minus free).

The maximum any vessel-side subsidy can remove is block A: 384.9 M€ in BAU, 545.5 M€ in NT0, smaller than the net gap in both scenarios (668.4 and 3,112.5 M€). Even a fully free vessel and terminal covers only 58% and 17% of the gap; the implied trigger discount on vessel capex (423% and 1,546%) is mathematically undefined. In other words, even if ships and infrastructure were given away for free, the maritime shift would still not be competitive against road, because the dominant cost block is the fuel production chain. Notably, net system investment is ≈0 (−10.8 M€): vessel capex is offset by avoided truck purchases, so a capital subsidy targets the one cost-neutral component of the bundle. In the Net-Zero branch the barrier is even larger, and it manifests indirectly. The direct marine-fuel chain (ammonia and LNG delivery, grey-ammonia synthesis, gas and hydrogen feed) adds +311.0 M€, non-negligible but not dominant, while the avoided freight assets and fuels save −1,349.8 M€. The dominant block is system rebalancing (+3,577.3 M€): under the binding emission cap, the grey-ammonia and LNG chains consume carbon space, and the system pays to re-abate elsewhere, accelerated passenger-car electrification (+1,370.3 M€) and costlier options in buildings, industry and power (+2,207.1 M€); 74% of the gap materialises outside the transport sector. The fuel-chain barrier of the BAU branch is therefore still the root cause, but in NT0 it is priced through the CO2 shadow price rather than as a direct line item, which is exactly why the mandate price rises from 5.32 to 14.95 M€2020 per Gtkm·yr.

4.3.3 Forcing ladders (comparisons C2–C4)

The forced runs measure the mandate as a dose–response curve (Figure 6, Table 6). In BAU, $\Delta C(\alpha) = \alpha \times 668.365$ M€ exactly (residuals < 1 k€), with HFO chosen endogenously at every α, a constant marginal cost of 5.32 M€2020 per Gtkm·yr. In NT0, the ladder is quasi-linear (marginal 775.6 → 779.7 M€ per 25% tranche, < 0.2% from linearity), ammonia-only at low shares with a 2040 LNG bridge entering at $\alpha \geq 75\%$, 14.95 M€2020 per Gtkm·yr.

Table 6: Forcing ladders (NT0 on the price-consistent baseline).

α	ΔC BAU [M€]	ΔC NT0 [M€]	NT0 marginal [M€/25%]	NT0 fuel
25%	167.09	775.64	775.6	AMM
50%	334.18	1,553.12	777.5	AMM
75%	501.27	2,332.76	779.6	AMM + LNG bridge
100%	668.37	3,112.47	779.7	AMM + LNG bridge

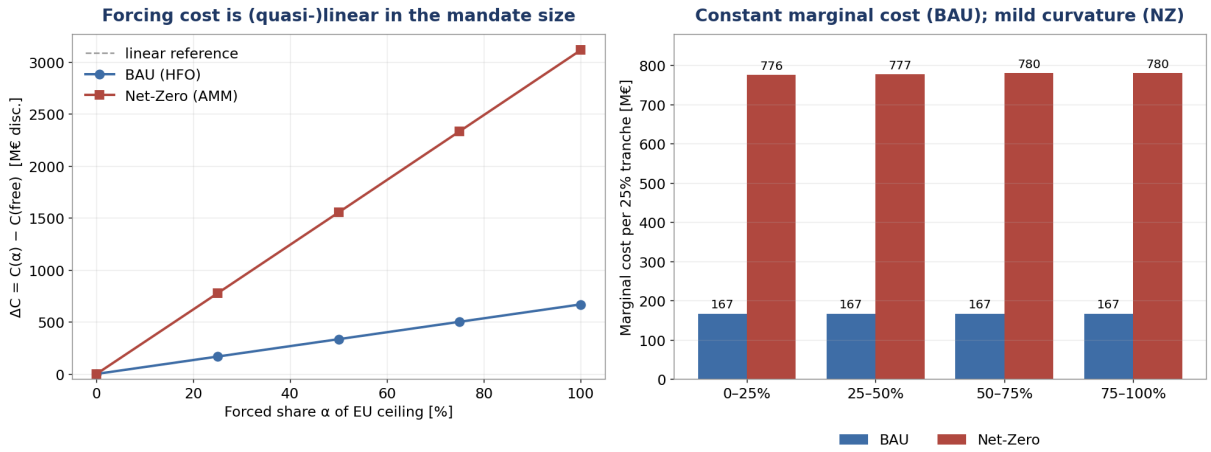


Figure 6: Forcing-cost ladders and marginal cost per tranche: exactly linear in BAU (5.32 M€/Gtkm·yr), quasi-linear in NT0 (14.95 M€/Gtkm·yr).

The scenario comparison is compact: same shifted quantity (8.338 Gtkm cumulative), cost ratio NT0/BAU = 4.66 on ΔC (80.2 vs 373.3 M€ per cumulative Gtkm) and 2.81 on the annualised incentive, decarbonising the road competitor nearly quintuples the cost of pulling freight off it.

4.4 Emissions impact: burden shifting, CCUS, and the diesel-train rationale

4.4.1 Full-horizon freight GWP_100 trajectories

The four reference scenarios diverge as expected, massively between BAU and NT0, marginally between free and forced (Table 7, Figure 7).

Table 7: Freight-sector GWP_100 trajectories [kt CO2-eq] across the four reference scenarios (perimeter: HTR/MTR/LCV road freight + freight rail + domestic navigation, shift twins included).

Scenario	2010	2020	2030	2040	2050
BAU free	44,984.68	33,131.87	29,679.44	25,294.67	24,649.65
BAU forced (FRZ)	44,984.68	33,131.87	30,065.95	25,786.24	25,252.85
NT0 free	44,984.68	30,620.49	26,071.30	11,767.96	3,358.00
NT0 forced (FRZ)	44,984.68	30,620.49	25,989.33	11,721.76	3,196.50

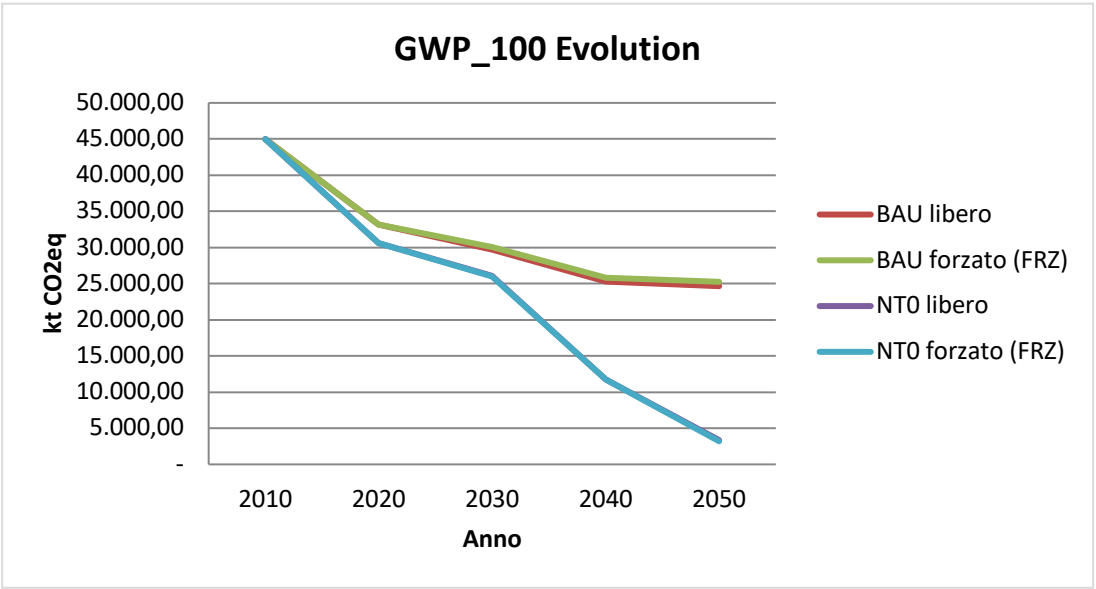


Figure 7: Evolution of GWP_100 by scenario: forcing the shift in BAU increases emissions at every period, while in NT0 the trajectories are pinned by the system-wide cap.

4.4.2 BAU: the mandate is carbon-positive

Forcing the shift in BAU increases emissions at every rung of the ladder (+125 → +650 kt CO2/yr system-wide), because the marine fuel used, HFO, is more polluting than the displaced road-transport mix. The displacement itself is smooth and split across two road powertrains: at 2050, D5 activity falls from 25.53 to 22.12 Bvkm between α = 0 and α = 100%, taken ~75% from full-hybrid trucks (14.11 → 11.53) and ~25% from natural-gas trucks (11.43 → 10.59). The mandate erodes the cleanest road vintages proportionally: from an environmental standpoint, modal shift makes sense only with clean marine fuels, or at least a significant clean-fuel share, in BAU it is a logistics policy, not a climate one.

4.4.3 Segment view and the free vs forced

Figure 8 decomposes freight GWP_100 by segment. Panel (a) contrasts the NT0 free (NT0_0) and forced (NT0_FRZ) runs: the 2050 totals are 3,358 vs 3,196 kt (−161 kt), a difference that falls almost entirely on rail (987 → 825 kt), while HTR_D5 and navigation read zero in both — confirming §4.4.1, where free and forced diverge only marginally under the binding cap. Panel (b) sets the two families side by side: the −21.3 Mt gulf between BAU and NT0 (24,650 vs 3,358 kt free at 2050) dwarfs any within-family forcing effect and is structural — in BAU every segment still emits (LCV ≈ 7.7, HTR ≈ 7.5, MTR ≈ 7.4, navigation ≈ 2.0 Mt), whereas NT0 zeroes road-freight HTR and navigation and confines the residual to the hard-to-abate carriers (LCV ≈ 2.1, rail ≈ 0.9, MTR ≈ 0.3 Mt). The two forcing directions are opposite in sign: in BAU the mandate is carbon-positive (+603 kt at 2050, dirtier HFO vessels, §4.4.2), in NT0 marginally carbon-negative (−161 kt), a pure reallocation effect: under a binding cap the total is effectively fixed, so the forcing merely shifts the residual budget onto rail (§4.4.4) rather than abating it.

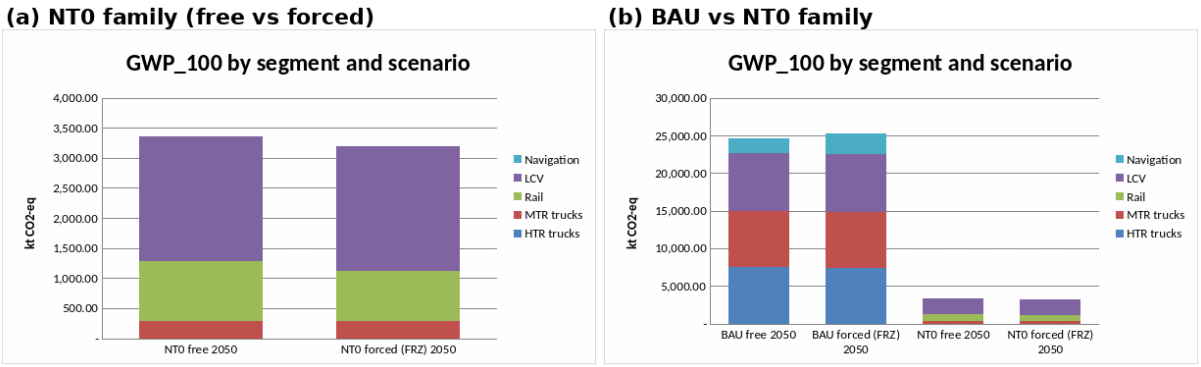


Figure 8: GWP_100 by freight segment and scenario: (a) NT0 family (free vs forced); (b) cross comparison of the BAU and NT0 families. In NT0 the residual emissions concentrate on rail while HTR_D5 and navigation are fully decarbonised.

The HTR_D5 segment is the one influenced by the modal shift. In both BAU_OFF and NT0_OFF the solver selects PHEV, a logical technology for long distances, economically sustainable and partially green. With the modal shift active, the required HTR_D5 fleet shrinks in ON_0 thanks to the entry of trains (and, in the forced runs, trains + navigation at the maximum): the solver then changes its choice, moving to full-hybrid in BAU and fuel-cell in NT0. The smaller truck fleet makes it affordable to install a more efficient and greener, but more expensive, technology per vehicle. In NT0 with modal shift ON, the PHEV share (the only residual tank-to-wheel emitter of the net-zero road freight fleet) is eliminated, making road freight fully zero-emission tank-to-wheel: the solver achieves complete decarbonisation of very-long-distance road transport only with the modal shift active. The underlying powertrain reallocation that drives this result — the switch away from PHEV once the modal shift frees fleet space — is analysed in §4.5 (Figure 12).

4.4.4 The diesel-train allocation

Symmetrically, in NT0_0 and NT0_FRZ the solver, having fully cut emissions on HTR_D5, reallocates part of them onto the rails, preferring to emit with diesel trains rather than with trucks. This produces the apparently paradoxical result that, in a net-zero system, the freight rail sub-sector has a higher carbon intensity than its BAU counterpart. This behaviour is fully rational for a cost-minimizing LP under a system-wide (not sectoral) emission constraint, through three concurrent mechanisms:

- **Marginal abatement cost arbitrage.** The GWP_100 constraint is aggregate: the model allocates the residual carbon budget where abatement is most expensive. The rail electrification premium (rolling stock, and implicitly infrastructure) exceeds the marginal abatement cost elsewhere, in particular that of CCUS negative emissions. It is cheaper to "pay for" diesel-rail emissions with low-cost removals than to electrify.
- **Competition on the electric carrier.** The 2050 system already stands at ~1,324 TWh (double the 2020 level), with PV and wind close to their potentials (Figure 10). Every PJ of electricity assigned to rail competes with end uses of higher marginal value; diesel rail frees electricity for uses where it substitutes carriers that are more expensive to decarbonise.
- **Effect of the forced modal shift.** The shift imposes an activity jump on rail (61 → 82 Gtkm) exceeding existing electric capacity: new vintages must be installed anyway, and at equal service the LP, with its winner-takes-all logic on new installations, selects the traction with the lowest total discounted cost, i.e. diesel.

The outcome (48 Gtkm diesel out of 82 at 2050, with hydrogen-fuelled road and ammonia-fuelled vessels fully decarbonised) is therefore a structural artefact of the aggregate constraint, not a technological judgement on electric rail. The power sector, it should be noted, is fully decarbonised in NT0 both free and forced (Figures 9–10).

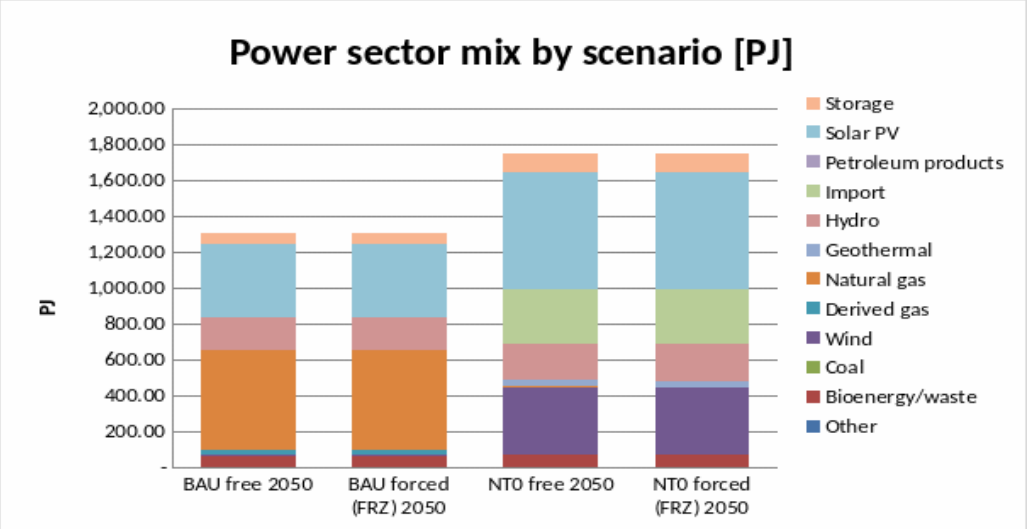


Figure 9: Power sector generation mix per scenario [PJ]: the NT0 branch is fully decarbonised in both the free and forced runs.

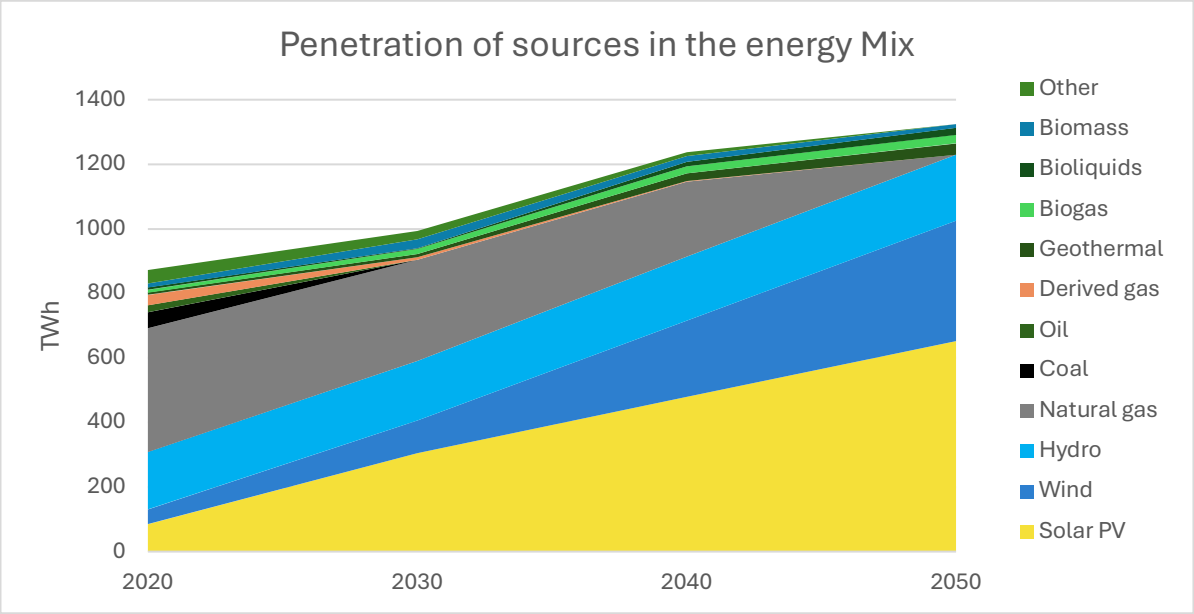


Figure 10: Penetration of sources in the 2050 electricity mix (~1,324 TWh): PV and wind close to their potentials, which makes the electric carrier a contended resource.

4.4.5 The FRZ_ELC controlled experiment

Two additional runs correct this allocation: NT0_FRZ_ELC_1 forces the shift trains to be fully electric; NT0_FRZ_ELC_2 forces the entire rail sector to be electric. Because the shifted Gtkm are identical in the three scenarios (the shift is imposed) and the 2050 electricity mix is identical (~1,324 TWh, renewables saturated), the ELC constraint operates exclusively on the traction choice: a clean controlled experiment with a single free variable. Four reportable results emerge:

- **The net effect of forced electrification is real but modest.** Freight emissions at 2050 fall from 3.20 (FRZ) to 2.97 (ELC_1) to 2.37 Mt (ELC_2): −26% on freight, but barely −0.4 Mt on the overall transport sector (~32 Mt), by then dominated by aviation and international maritime. The constraint acts on a small fraction of the hard-to-abate residual.
- **The system compensates rather than abates.** The increase in the power sector (0.61 → 0.72 Mt) and in industry (+0.36 Mt in ELC_2, driven by electricity and carrier demand) erodes part of the gain; the net system benefit is smaller than the sectoral one, a classic reallocation effect under an aggregate constraint, quantifiable only through multi-sector comparison (Figure 11, Table 8).
- **Activity and generation mix are invariant,** isolating the traction variable, a point worth stating explicitly in any methodology section.
- **The FRZ diesel-rail result is LP rationality, not an anomaly.** ELC_1/ELC_2 demonstrate that correcting the artefact costs the system very little (marginal objective variations): an argument in favour of sectoral technology constraints whenever the LP output produces realistically implausible allocations.

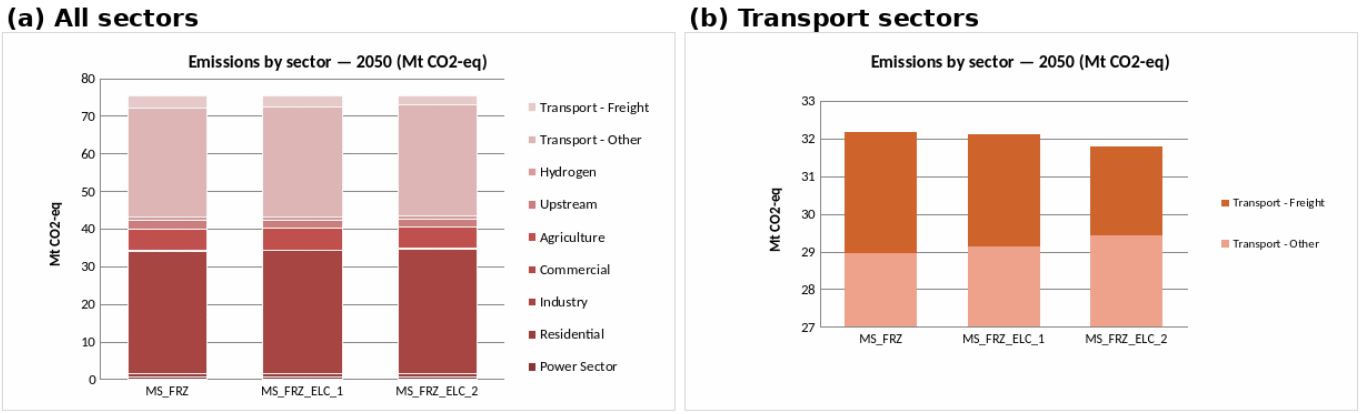


Figure 11: Sectoral GWP_100 decomposition at 2050 for the three forced NT0 variants: (a) all sectors; (b) transport split (other vs freight). The forcing changes the "where" of emissions more than the "how much".

Table 8: Sectoral GWP_100 decomposition of the three forced NT0 variants at 2050 [Mt CO2-eq].

Sector	NT0_FRZ100	NT0_FRZ_ELC_1	NT0_FRZ_ELC_2
Power sector	0.607	0.620	0.724
Residential	0.880	0.880	0.767
Industry	32.698	32.784	33.055
Commercial	0.184	0.198	0.290
Agriculture	5.689	5.689	5.689
Upstream	2.237	2.224	2.182
Hydrogen	0.852	0.845	0.829
Transport, other	28.970	29.133	29.418
Transport, freight	3.196	2.969	2.371
Total (positive sectors)	75.313	75.342	75.325
CCUS (sink)	−43.722	−43.731	−43.664

The synthesis for the discussion: the forcing changes the "where" of emissions more than the "how much", and this, under net-zero with CCUS available, is exactly the expected behaviour of a system-optimum model.

4.4.6 The ammonia chain and burden shifting

In NT0, the maritime shift (whether forced or, hypothetically, spontaneous) is a direct substitution between HFO and ammonia. The ammonia production chain, reconstructed from the solved database of the forced Net-Zero run (NT0 FRZ100), has three stages: (1) Synthesis (IND_CH_AMM_NGASR_N) , steam reforming of natural gas, $\eta = 0.35$ (vintage 2030+), with gas as fuel (100.2 PJ at 2050) and feedstock (125.3 PJ) plus auxiliary electricity (2.3 PJ), producing 6.4 Mt NH3 and a non-separable chemical co-product (IND_CH_SB, 73.3 Mt); the optimum selects exclusively the grey route , neither CCS, nor electrolysis, nor biomass gasification proves competitive. (2) Fuel technology (TRA_FT_AMM_IND_N), mass-to-energy conversion at 18.6 PJ/Mt (LHV), variable cost 22.28 M€/PJ, no emitting transformation. (3) End use (TRA_NAV_DOM_AMM_ICE_N + SHIFT twin), marine ICE engines, 153.8 M€/Gtkm investment, covering the entire domestic navigation (12.7 Gtkm) at 2050. Tables 9–10 quantify the chain.

Table 9: The ammonia chain (from the solved NT0 FRZ100 database).

Item	2030	2040	2050
Natural gas input , fuel IND_NGA (PJ)	37.04	62.85	100.21
Natural gas input , feedstock FS_NGA (PJ)	46.30	78.56	125.26
Electricity input IND_CH_EC (PJ)	0.84	1.43	2.28
Total NH3 production , SMR + coal gasif. (Mt)	3.07	4.00	6.38
Co-product output IND_CH_SB (Mt)	26.67	45.99	73.33
Ammonia to transport , FT input (Mt)	2.56	3.49	5.85
TRA_AMM delivered (PJ) [= Mt × 18.6]	47.62	64.91	108.81
Ammonia-fuelled DOM naval activity (Gtkm)	1.62	1.57	11.69

Table 10: GWP_100 by chain step: emissions entirely concentrated in synthesis [kt CO2-eq].

Emissions by step (GWP_100)	2030	2040	2050
Step 1 , SMR: realized GWP_100	2,100.3	3,546.7	5,655.0
, of which CO2 (kt)	2,087.9	3,525.7	5,621.6
, of which CH4 (kt, ×0.021 in GWP)	40.9	69.1	110.2
, of which N2O (kt, ×0.31 in GWP)	37.2	62.9	100.2
Step 2 , Fuel technology	0.0	0.0	0.0
Step 3 , Marine ICE combustion	0.0	0.0	0.0
Intensity: GWP per Mt NH3+SB produced (kt/Mt)	70.6	70.9	70.9

Emissions are entirely concentrated in Stage 1: EF ≈ 238.7 kt CO2/PJ of input gas (plus fugitive CH4 and N2O), for a total of 2.1 → 5.7 Mt CO2-eq between 2030 and 2050, growing with naval demand. Final combustion is zero-carbon (carbon-free energy carrier), with one relevant modelling limitation: N2O slip from ammonia ICEs is not parameterised, whereas in reality N2O emissions (GWP ~273) are a known issue of this technology. The critical point for the discussion: in accounting terms, emissions migrate from the TRA sector to the IND sector, the ship is "clean", the system is not. This is burden shifting, not abatement; under the net-zero constraint the residual load is compensated by the CCUS compartment (−43.7 Mt at 2050).

4.5 Technology composition and the role of distance classes

Figure 12 shows the technology shares of the HTR_D5 segment, the market directly contended by the modal shift, across the six reference runs, while Figure 13 aggregates the 2050 fleet composition of the three vehicle categories across all scenarios. The technological divide between BAU and NT0 is clear-cut and driven by the emission cap and ETS pricing; within each family, the modal shift reshapes the D5 powertrain choice as discussed in §4.4.3.

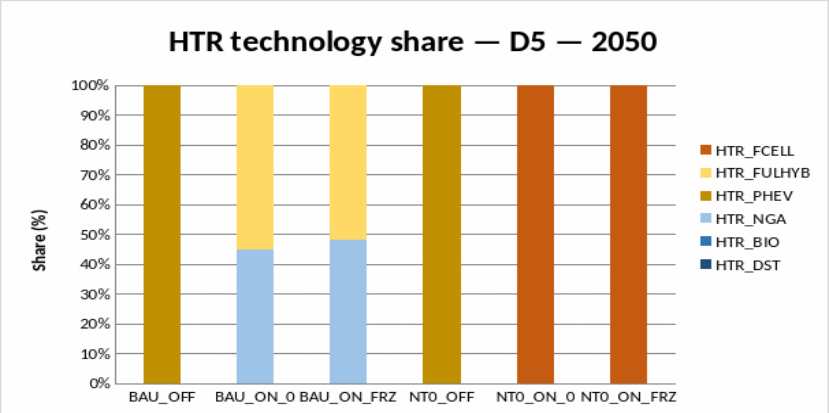


Figure 12: Technology shares of HTR, D5 at 2050 across the six reference runs: PHEV in the OFF runs is replaced by full-hybrid (BAU) and fuel-cell (NT0) once the modal shift is active. Powertrain codes: DST diesel, BIO biofuel, GSL gasoline, NGA natural gas, PHEV plug-in hybrid, FULHYB full-hybrid, ELC battery-electric, FCELL hydrogen fuel cell.

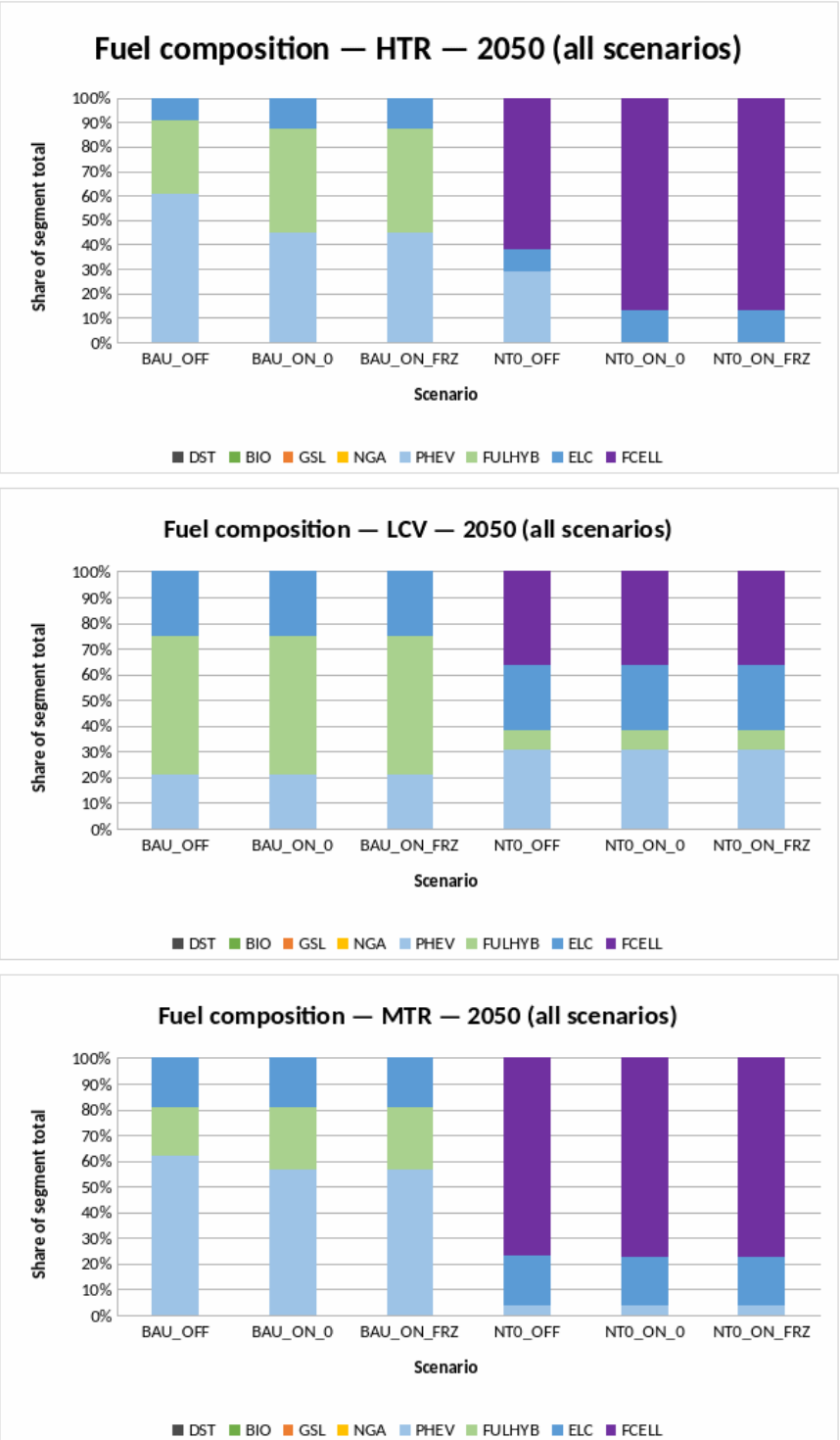


Figure 13: Fleet-wide 2050 fuel composition across all scenarios (top to bottom: HTR, MTR, LCV). The BAU/NT0 divide dominates; the modal shift affects only the long-haul heavy segment.

4.5.1 Heavy trucks (HTR) by distance class

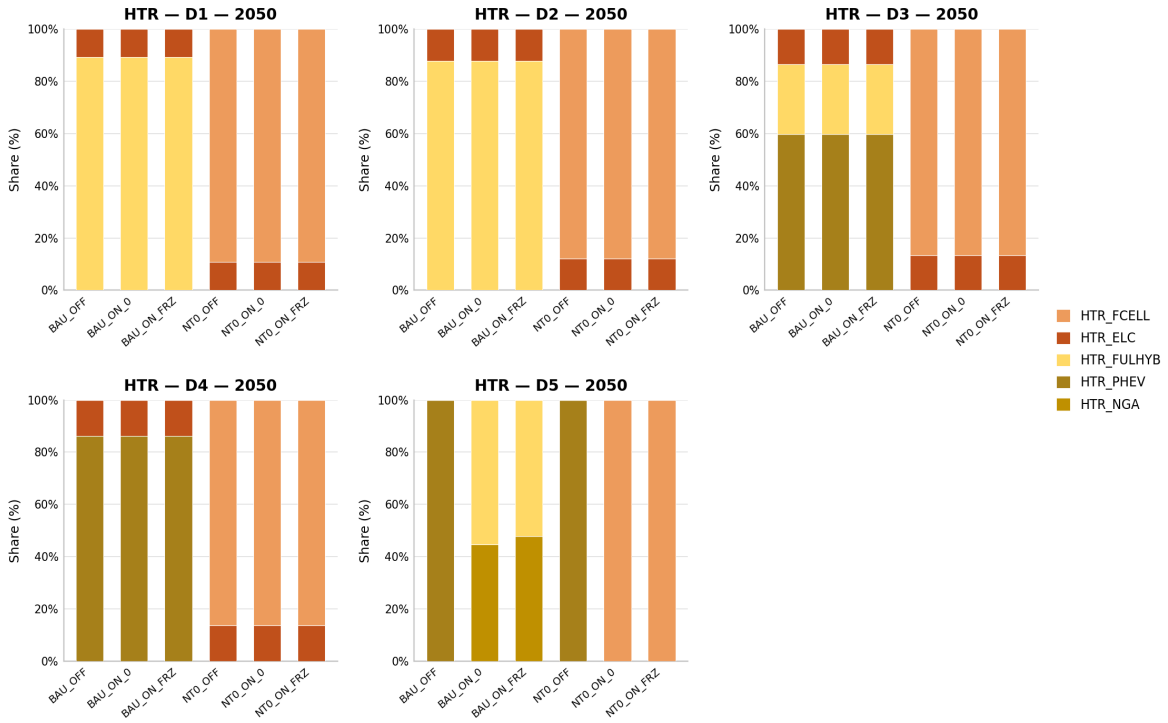


Figure 14: Technology shares of heavy trucks (HTR) at 2050 by distance class (D1–D5) across the six reference runs.

The distance-class resolution proves its analytical value here, and the divide runs along scenarios more than along distances. In the BAU family, D1–D2 rely on a full-hybrid/battery-electric duopoly (89%/11% at 2050 in D1), with PHEV taking over the mid-to-long classes (60% in D3, 86% in D4); fuel cells never appear at any distance. In the NT0 family the picture is inverted: hydrogen fuel-cell trucks dominate every class from D1 to D4 with a remarkably flat share (86–89%), complemented by battery-electric vintages (11–14%), and this D1–D4 mix is identical across all six runs, the carbon cap, not the haul length, selects the powertrain. D5 is the only class where the runs diverge, because it is the only market contended by the modal shift: in NT0_OFF the residual demand is served entirely by PHEV, while in every modal-shift run the residual road activity pivots to a 100% fuel-cell fleet; in BAU, D5 moves from 100% PHEV (OFF) to a full-hybrid/natural-gas pairing (~52%/48%) once the architecture is active , the two vintages the maritime mandate then erodes (~75%/25%, cf. §4.4.2).

4.5.2 Medium trucks (MTR) by distance class

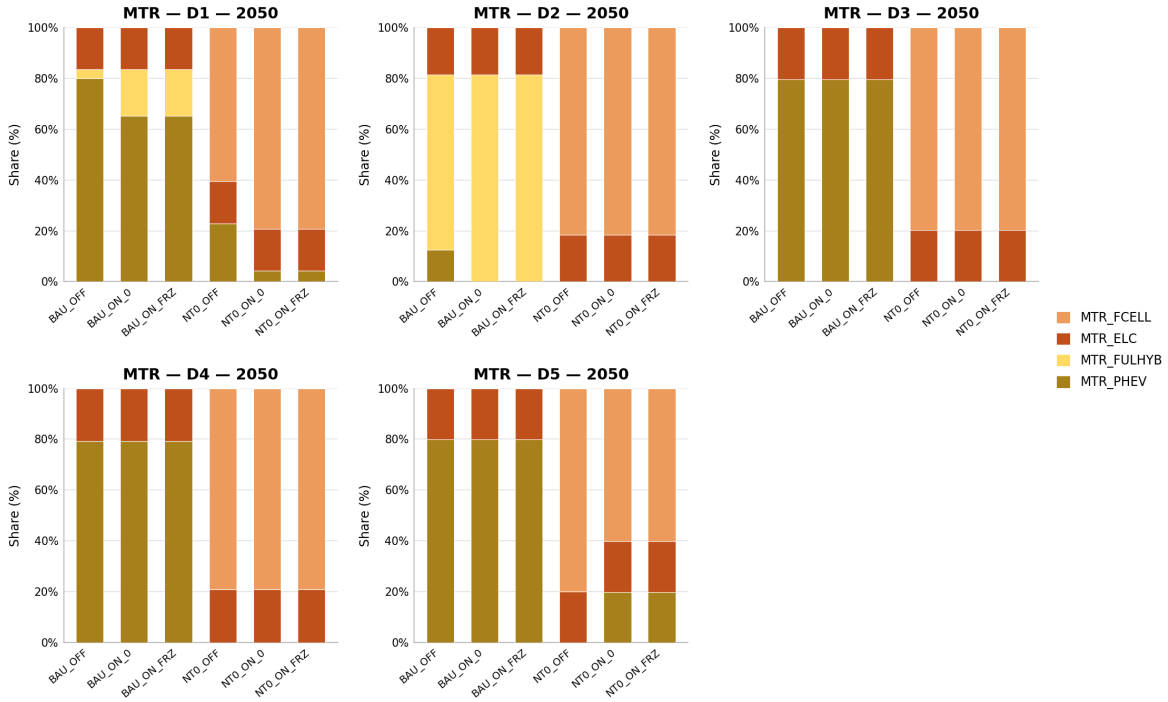


Figure 15: Technology shares of medium trucks (MTR) at 2050 by distance class (D1–D5) across the six reference runs.

Medium trucks replicate the heavy-truck pattern at lower stakes. In NT0, fuel cells are present from the very first distance class (61–79% already in D1) and hold a stable 79–82% share across D2–D4, complemented by battery-electric vintages (16–21%); in BAU the mix is a stable combination of PHEV and battery-electric (with full-hybrid in the shorter classes), essentially invariant from D2 to D5. What distinguishes the distance classes is therefore not the entry point of hydrogen but the behaviour at the boundaries: D2–D4 are identical across all six runs, while D5 diverges in NT0, the modal-shift runs admit a 20% PHEV share at 2050 (0.43 Bvkm displacing fuel-cell activity) that never appears in NT0_OFF. This is a knock-on effect of the shift in the adjacent HTR_D5 market: the combustion allowance released there can be reallocated under the common cap, and MTR_D5 is where it buys the largest cost saving (§4.5.4). Since MTR does not participate directly in the D5 modal-shift market, this indirect coupling is itself informative: the modal-shift layer perturbs the market it was built for, plus the neighbouring segments that trade emission headroom with it.

4.5.3 Light commercial vehicles (LCV) by distance class

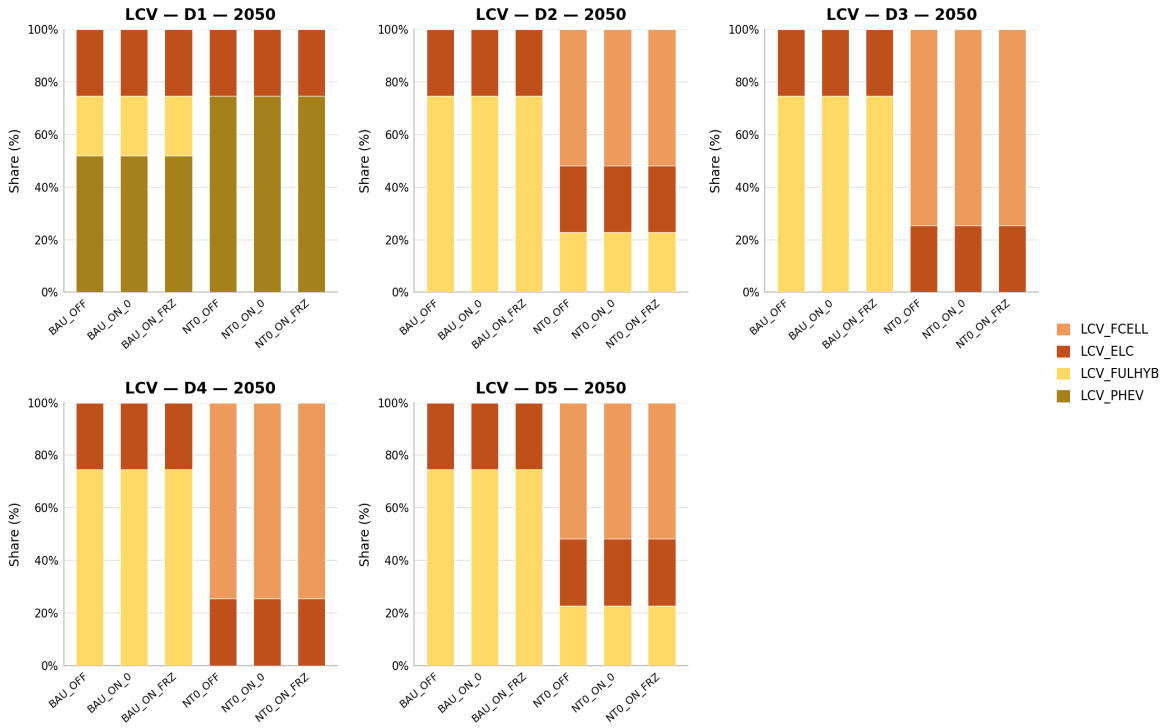


Figure 16: Technology shares of light commercial vehicles (LCV) at 2050 by distance class (D1–D5) across the six reference runs.

LCVs display the sharpest D1 signature of the three categories: D1 , which alone carries 37% of LCV activity (Table 1) , is the plug-in-hybrid stronghold in NT0 (75% PHEV, 25% battery-electric at 2050), consistently with the urban last-mile duty cycle in which short daily mileage and depot charging let a PHEV operate essentially in electric mode at a lower capital cost than a pure BEV. From D2 onwards the net-zero mix switches to hydrogen: fuel cells hold 52–75% of every class from D2 to D5, complemented by a stable 25% battery-electric share (with full-hybrid vintages surviving in D2 and D5); in BAU, D2–D5 settle on an invariant full-hybrid/battery-electric split (75%/25%). As for MTR, the LCV mixes are identical across the six runs , the category never touches the D5 modal-shift market. Taken together, Figures 14–16 confirm the central methodological claim of Part A: the distance classes discriminate the powertrains by their intrinsic characteristics (range, refuelling time, energy efficiency), a resolution the original aggregate representation could not provide by construction.

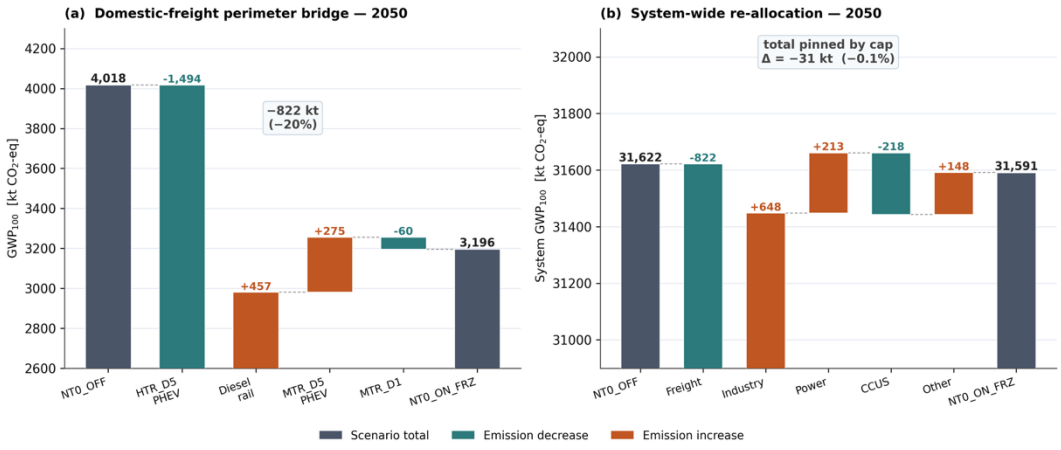
4.5.4 NT0_OFF vs NT0_ON_FRZ: the architecture ON/OFF comparison

The two architecture-ON versus OFF comparisons are asymmetric. In BAU the intermodal architecture barely moves freight emissions, the 2050 trajectories differ by less than ±0.3%, since the endogenous rail shift substitutes near-equivalent road emissions, so there the mandate is a logistics policy, not a climate one. In NT0, by contrast, the architecture is one of the levers the optimiser actually uses to meet the cap; the OFF/ON pair below isolates how.

Comparing the free Net-Zero run without the intermodal architecture (NT0_OFF) with the fully forced modal-shift run (NT0_ON_FRZ) provides a controlled experiment on the allocation, rather than the amount, of emissions. Since NT0_OFF represents road freight in vehicle-km while the modal-shift database works in tonne-km, the comparison is drawn on physical emissions and technology shares, which are representation-independent. At the system level, total GWP_100 at 2050 is pinned by the common cap: 31,622 kt in NT0_OFF versus 31,591 kt in NT0_ON_FRZ, a 0.1% difference. Within the domestic freight perimeter, however, emissions fall from 4,018 to 3,197 kt (−822 kt), and the delta closes exactly on four terms: −1,494.5 kt from the disappearance of the HTR_D5 PHEV fleet, +456.9 kt from diesel rail traction (the diesel shift twin alone emits 500.3 kt, partially offset by base-rail rebalancing), +275.4 kt from a new MTR_D5 PHEV vintage and −59.6 kt from MTR_D1. The budget freed inside freight is re-spent where a tonne of CO2-eq buys the largest system cost saving: industry retains +648 kt of combustion, the power sector adds +213 kt serving the electric rail traction, and CCUS deepens net capture by 218 kt, with minor offsets in the residential, commercial and upstream sectors. The modal shift does not abate a single net tonne under a binding cap — it relocates abatement to where it is cheapest.

The fleet history of HTR_D5 behind the −1,494.5 kt differs more than the 2050 snapshot suggests. In NT0_OFF the segment converts to PHEV as early as 2030 and stays 100% PHEV to the horizon: with the whole D5 demand on road, the plug-in architecture is the least-cost compromise between the shrinking combustion allowance and the electrification requirement. In NT0_ON_FRZ the same segment remains diesel-fuelled through 2030, bridges through natural gas and the first fuel-cell vintages in 2040, and lands on a 100% fuel-cell residual fleet in 2050: once rail absorbs the shiftable share, the residual road demand no longer justifies holding a combustion allowance in this segment, whose marginal value is higher elsewhere in the capped system.

The knock-on effects are two, and both instructive. First, the PHEV expelled from HTR_D5 reappears next door: MTR_D5, identical between the two runs through 2040, admits at 2050 a 0.43 Bvkm PHEV vintage (20% of the class) displacing fuel-cell activity. Under a binding common cap the linear program equalises marginal abatement costs across segments, so the allowance released by heavy trucks migrates to the adjacent market where keeping combustion is cheapest — the same duality logic that prices the maritime mandate in §4.4. Second, diesel traction survives inside the rail system precisely because the shift makes it valuable: the diesel shift twin carries 29.2 Gtkm and 500.3 kt at 2050, the winner-takes-all outcome expected from an LP without integrality restrictions. Everything else is untouched: the D1–D4 mixes of all three truck categories are strictly identical between the two runs, confirming that the intermodal architecture perturbs only the D5 market it was built for, plus the sectors that trade emission headroom with it.



4.6 MGA: the near-optimal space

Twenty HSJ (hop-slip-jump) iterations at 1% slack (45.5 bn€ of tolerance) on BAU_0: additional domestic navigation appears in none of the 20 maximally different near-optimal solutions; the near-optimal space is rich only along the D5 road-powertrain dimension (NGA, LPG, FCELL, ELC quasi-interchangeable within 1% of system cost). Notably, the MGA iterations also show a collapse of the rail shift , but this is structural, not economic: the activity-weighted HSJ objective counts the same shifted Gtkm on the three technologies of the chain (twin, terminal, linker) and therefore suppresses intermodal chains by accounting construction. The rail-shift collapse in the MGA iterations is an artefact of the alternatives-generation objective, not an economic result.

4.7 Monte Carlo sensitivity on intermodal terminal costs

To test the robustness of the modal allocation against the least certain parameters of the architecture, a Monte Carlo campaign was run on the cost parameters (CostInvest, CostFixed, CostVariable) of the two intermodal terminals in the BAU free run. Ten samples were drawn from a normal distribution with a standard deviation of 50% of the reference value, applied with a linear ramp over vintages and periods after 2030 (the 2030 vintage is therefore unperturbed by construction); each perturbed database was solved and restored, and every sampled value was archived in dedicated MonteCarlo audit tables, consistently with the INSERT-only audit-trail discipline of the rest of the database. The sampled terminal costs span from −92% to +124% of the reference values: the NAV terminal CostInvest at 2050 ranges between 0.71 and 20.76 M€/Gtkm against a reference of 9.29.

The modal allocation is invariant across all ten samples. Rail shift remains at its EU ceiling in every period (26.71, 34.50 and 42.55 Gtkm in 2030, 2040 and 2050) to the third decimal digit; maritime shift remains exactly zero; the 2050 twin mix is 100% electric in every run. The total system cost varies between 4,551,974 and 4,552,030 M€ , a 55.7 M€ spread that confirms the perturbations do reach the objective function. The most informative draw is the one pricing the NAV terminal at 0.71 M€/Gtkm: a de facto subsidy above 90% on the intermodal cost component, beyond the entire S0–S150 ladder, which still activates no maritime shift whatsoever.

The result is the signature of a robust corner solution: the optimal basis is unchanged by the perturbation, which is entirely absorbed by the dual variables. It certifies numerically, with a method independent from the cost-bridge decomposition of Section 4.3.2, the two structural findings of this chapter: the rail ceiling constraint is active with a cost margin far wider than a ±50% terminal-cost uncertainty, and the maritime competitive gap lies outside the transhipment cost component altogether. With ten samples the campaign is not a distribution estimate; it is a basis-invariance check, for which the relevant quantity is the width of the sampled range, not the sample size.

5. Discussion and limitations

The results support a clear policy reading: the two EU modal-shift channels are governed by different binding constraints. Rail is limited by infrastructure capacity, its ceiling saturates spontaneously, while domestic navigation is limited by the economics of the marine-fuel supply chain, a block that no sector-internal financial instrument can reach. The cost bridge, the upstream energy accounting and the forcing ladders triangulate the same conclusion from three independent output tables, which is the strongest internal-consistency argument the model can offer. Under net-zero, the aggregate emission constraint relocates residual freight emissions (diesel rail, grey ammonia) rather than removing them: a system-optimal behaviour that sectoral technology constraints (FRZ_ELC) can correct at negligible system cost.
The literature on endogenous modal shift in energy system optimization models has followed a different route: Salvucci et al. [9] and Salvucci, Gargiulo and Karlsson [10] endogenise modal competition in TIMES through elasticities of substitution, letting modal demands adjust to shadow-price variations across distance classes. The present architecture is complementary rather than alternative: instead of a behavioural response calibrated on literature elasticities, it makes the modes physically compete on a common tonne-kilometre market under explicit infrastructure ceilings, so the result is a pure techno-economic potential. The conclusions nonetheless converge. Salvucci et al. [10] find rail-based freight shift to be cost-effective for Scandinavian decarbonisation, consistent with the no-regret rail result found here, while Pedinotti-Castelle et al. [11] show, on the NATEM model, that endogenous mechanisms deliver only limited shift and that disruptive modal change must be imposed exogenously: precisely the conclusion this work reaches for domestic navigation, where the mandate (or upstream fuel-chain decarbonisation) is the only effective instrument. The distinctive contribution here is the decomposition of why the endogenous channel fails for the maritime mode, the cost bridge locates the barrier in the marine-fuel chain, outside the reach of any sector-internal subsidy.
The main limitations of the study are the following.
Homogeneous freight demand. The aggregated TRA_FRG commodity treats all D5 tonne-kilometres as perfectly substitutable at fixed modal load factors: cargo heterogeneity (perishables, just-in-time logistics, high-value goods) is not represented, so the technical shift potential is an upper bound.
Point-estimate terminal parameters. The intermodality penalties (0.87 rail, 0.78 sea) and terminal costs are point estimates. The Monte Carlo campaign of Section 4.7 shows, however, that the modal allocation is invariant to terminal-cost perturbations of ±50% (1σ); the residual concern is confined to the intermodality penalties, whose efficiency-based implementation on a free source commodity makes them economically inert by construction, migrating them to the twins' fuel and access requirements is the natural refinement of this architecture. Since the dominant cost block is the marine-fuel chain, plausible variations change the magnitude of the gap but not the sign of the incentive result.
Scope of the EU target. The modal-shift ceiling is applied to the D5 (>500 km) market only, consistent with the >300 km policy perimeter but conservative on the D4 band.
Perfect foresight. Anticipated investment decisions structurally favour the no-regret rail shift; a myopic formulation could delay ceiling saturation.
Emission parameterisation. N2O slip from ammonia marine ICEs (GWP ~273) is not modelled; the dual-fuel HFO+MET twin is excluded by the additive-split limitation and documented in the source.
MGA objective. The activity-weighted HSJ formulation triple-counts intermodal chains (twin, terminal, linker) and suppresses them by construction: MGA results are informative on the road-powertrain space, not on the modal dimension.

6. Conclusion

- **The project's core deliverable is the database:** ~9,500 lines of INSERT-only, self-documenting SQL that add distance-class resolution (D1–D5) and a validated endogenous modal-shift architecture (aggregated Gtkm demand, zero-cost linkers, terminals with physical intermodality penalties, blended shift twins via TechInputSplit, disjoint BASE/SHIFT groups with BAU floors and EU ceilings) to TEMOA-Italy, one patch reused unchanged across 14 runs, two ELC diagnostic variants and 20 MGA iterations.
- **Rail modal shift is a no-regret measure:** the EU ceiling is saturated in every free run with zero incentive; the binding constraint is infrastructure, not economics. Under the NT0 cap it contributes −651 kt CO2 in 2050 versus the OFF architecture.
- **Maritime modal shift cannot be bought from inside the transport sector:** the incentive runs are identical to the free runs on every metric, and the accounting proof shows why, the gap (668/3,112 M€) exceeds the whole subsidizable chain (385/546 M€); the unreachable remainder is the fuel chain, confirmed physically upstream (crude +20 PJ, refinery +191 PJ). A ten-sample Monte Carlo on terminal costs (σ = 50%, draws from −92% to +124% of reference) leaves the allocation of every run unchanged, rail at ceiling, navigation at zero, certifying both findings as robust corner solutions.
- **A mandate has an exact price:** 5.32 (BAU, exactly linear) and 14.95 (NT0, quasi-linear, LNG bridge only at α ≥ 75%) M€2020 per Gtkm·yr, 0.67 to 3.1 bn€ discounted for the full EU target; ratio NT0/BAU = 4.66 on total cost.
- **In BAU the mandate is carbon-positive at every rung,** because HFO vessels displace the cleanest road vintages (~75% full-hybrid, ~25% gas): a logistics policy, not a climate one. Environmental value requires clean marine fuels.
- **Under net-zero, the model relocates rather than removes residual freight emissions:** diesel trains and grey-ammonia synthesis are rational LP allocations under a system-wide cap with CCUS available (the forcing changes the "where" more than the "how much"). The FRZ_ELC controlled experiments show that correcting the diesel-rail artefact is cheap for the system (freight 3.20 → 2.37 Mt; −0.4 Mt net on ~32 Mt of transport), supporting the use of sectoral technology constraints when aggregate-constraint LP allocations are realistically implausible. The ammonia route, as selected by the optimum, is burden shifting from TRA to IND, not abatement , and omits N2O slip from marine ICEs, a known real-world issue.
- **Policy synthesis:** EU-style naval modal-shift targets in Italy require regulatory mandates or upstream decarbonisation of marine fuels; sector-internal financial incentives, of any intensity, are the wrong instrument by construction.

Method note , *all figures derive from the solved databases (14-run tree + 2 ELC variants) and 20 MGA iterations, audited on Output_Objective, Output_VFlow_In/Out, Output_Costs, Output_Emissions; discounting 5% to 2007.*

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Workload table

Project Phases	Project parts	F.Cinquini	C.Chiapparoli
Implementation	Model development and scenarios definition	*50%*	*50%*
	Methodology definition	*50%*	*50%*
	Analysis of results and post-processing	*65%*	*35%*
Report preparation	Model and scenarios description	*60%*	*40%*
	Description of the adopted methodology	*60%*	*40%*
	Description of the results	*50%*	*50%*
Presentation		*50%*	*50%*